

Kiernan X. Jennings

Ph.D. Candidate

Department of Chemical and Biological Engineering, University of Wisconsin-Madison

Email: kxjennings@wisc.edu | Cell: +1 860 259 7853 | LinkedIn: [kiernanjennings](#)

Website: <https://kiernanj.github.io>

EDUCATION

University of Wisconsin-Madison

2024 - Present

3rd Year Ph.D. Candidate in Chemical Engineering, Minor in Industrial and Systems Engineering

Thesis Advisors: Victor M. Zavala and Styliani Avraamidou

GPA: 3.77

University of Connecticut

2020 - 2024

B.S. in Chemical Engineering, *Cum Laude*

GPA: 3.88

RESEARCH INTERESTS

Large-scale optimization; decomposition algorithms; energy systems

EXPERIENCE

University of Wisconsin-Madison

2024 - Present

Graduate Research Assistant

Advisors: Victor M. Zavala and Styliani Avraamidou

- Developed a multi-scale MILP framework that couples demand-response operation of electrolyzers in day-ahead and real-time markets with usage-based degradation and stack replacement scheduling.
- Designed an aggregate-informed Benders decomposition algorithm. On 40-year instances with up to 5.6M binary variables, where monolithic solves run out of memory, it cut final optimality gaps from 85% to under 5%.
- Building PlasmoSurrrogates.jl, a Julia framework built on PlasmO.jl for decomposing optimization models that combine ML surrogates with algebraic models.

PSOR Lab, University of Connecticut

2023 - 2024

Undergraduate Researcher

Advisor: Matthew D. Stuber

- Developed acausal model of wastewater treatment and anaerobic digestion using ModelingToolkit.jl and implemented models for EAGO, an open-source global optimization program written in Julia
- Developed advanced discrete event handling algorithms for PID controller in ModelingToolkit.jl

RESEARCH OUTPUT

Publication Preprints/Under Review

- **K.X. Jennings**, V.M. Zavala, and S. Avraamidou. A Multi-Scale Optimization Framework for Grid-Integrated Electrolysis. *Submitted to Computers & Chemical Engineering. arXiv preprint arXiv:2607.09512* (2026) [\[link\]](#)

Conference Publications (Peer Reviewed)

- **K.X. Jennings**, V.M. Zavala, and S. Avraamidou. Decomposing a Multi-Scale Optimization Framework for Grid-Integrated Electrolysis using Aggregate-Informed Benders. In 2027 Foundations of Computer Aided Process Operations - Chemical Process Control (FOCAPO-CPC) [\[link\]](#)

Oral Presentations

- P. Brahmabhatt, **K.X. Jennings**, V.M. Zavala and S. Avraamidou. Decision-Focused Surrogate Modeling for Mixed-Integer Nonlinear Programs Via Neural Differentiable Optimization Layers. *2026 AIChE Annual Meeting*, Minneapolis, MN. [\[link\]](#)
- P. Brahmabhatt, **K.X. Jennings**, A. McCullough, V.M. Zavala, and S. Avraamidou. Multi-Market Bidding and Demand Response for Redox Reservoir-Based Electrochemical Devices Using Two-Stage Stochastic Optimization. *2025 AIChE Annual Meeting*, Boston, MA. [\[link\]](#)

Poster Presentations

- **K.X. Jennings**, S. Avraamidou, and V.M. Zavala. Multi-Scale Optimization and Decomposition Framework for Grid-Integrated Electrolysis Systems. *TWCCC 2026*, Madison, WI. [\[link\]](#)
- **K.X. Jennings**, S. Avraamidou, and V.M. Zavala. Multi-Scale Optimization for Grid-Integrated Electrolysis: Demand Response Operation with Usage-Based Degradation and Stack Replacement Scheduling. *Hougen PSE Symposium 2026*, Madison, WI.
- **K.X. Jennings**, S. Avraamidou, and V.M. Zavala. Multi-Scale Optimization for Capacity Planning and Operations of Electrolysis Systems. *TWCCC 2025*, Madison, WI. [\[link\]](#)
- **K.X. Jennings**, S. Avraamidou, and V.M. Zavala. Multi-Scale Optimization for Evaluating Electrolyzer Operation Degradation in Electricity Markets. *Power Grid-Informed Electrochemistry Conference 2025*, Madison, WI. [\[link\]](#)

MENTORING & TEACHING EXPERIENCE

Training

- UW-Madison Delta Program Research Mentor Fall 2026

Advised Students

- Emilia Veum May 2026-Present
Project: Carbon Aware Grid-Integrated Electrolysis Optimization
- Aayana Roy Jun 2025-May 2026
Project: Capacity Expansion Modeling of UW-Madison Campus

Teaching Experience

- Teaching Assistant, Process Design Senior Capstone, UW-Madison Fall 2025, Spring 2026

SKILLS & COURSES TAKEN

Skills

- Optimization (JuMP.jl, Plasmol.jl)
- Control Systems (Simulink)
- Process Simulation (AspenPlus)
- Programming (Julia, Matlab, Python, C)

Courses

- CBE 660 (Advanced Mathematics)
- CBE 710 (Advanced Thermodynamics)
- CBE 735 (Kinetics and Catalysis)
- CBE 750 (Advanced Chemical Process Synthesis and Optimization)
- CBE 781 (Biological Engineering)
- ISyE 524 (Introduction to Optimization)
- ISyE 726 (Nonlinear Optimization I)
- ISyE 823 (Mixed-Integer Nonlinear Optimization)

AWARDS & SERVICE

Outreach Services

- | | |
|--|--------------|
| - University Engineering Expo Volunteer | 2024-Present |
| - Circular Economies Systems Group Climate & Inclusion Officer | 2024-Present |
| - Swim Across the Sound Group Leader | 2022-2024 |
| - HuskyTHON Student Group Leader | 2021-2024 |

Academic Awards

- | | |
|---|-----------|
| - Sunny and Po Lo Graduate Fellowship Award | 2024-2025 |
| - Homer Babbidge Scholar | 2021-2022 |
| - Dean's List | 2021-2024 |
| - Academic Excellence Award | 2020-2024 |